1

Sequential backward selection starts with a full model and removes features until typically there are no more features to be removed (either reached threshold ie k or a null model) or there are no improvements with removing any more features. In the case of this question however, it appears we will always remove features until we reach our threshold k.

To find how many total models trained lets look at the backward selection step by step, the first of which is training a full model. Then the second step is training models on one less feature than our current model. In the case of the second step that would be m different models of size $m-1$, but for the third it would be $m-1$. We repeat this retraining step until we reach models with k features, in which we would have to train $k+1$ of them.

We can create a simple formula for determining the total number of models trained, with 2 different cases:

Case 1: $m-k$ is even

$1+\frac{m-k}{2}(m+k+1)$

Case 2: $m-k$ is odd

$1+m+\frac{m-1-k}{2}(m+k)$

We can verify this formula with the example $m=10, k=5$, which our formula says $1+10+30=41$ models will need to be trained:

1 full model

10 models with 9 features

9 models with 8 features

8 models with 7 features

7 models with 6 features

6 models with 5 features

Total: 41

Just to test both cases we can also test on $m=9, k=5$, which our formula says $1+30=31$ which lines up with the info found above (we simply remove the models with 9 features from the work above to determine the total number of models trained with $m=9, k=5$).

2

Let $S(\lambda)$ represent the sorted list of eigenvalues that are highest to lowest. This question is asking us for k eigenvalues such that $\sum_{i=1}^{k} \frac{S(\lambda)_i}{\sum_{j=1}^{m} \lambda_j} = 0.7$

> First it is important to determine $\sum_{j=1}^{m} \lambda_j$ which in this case is $0.22 + 0.9 + 0.1 + 0.07 + 0.08 + 0.03 + 0.8 + 0.003 + 0.19 = 2.393$
>
> Then we simply sum eigenvalues in decreasing order until this total reaches 0.7. Thus, the largest eigenvalue is 0.9 which accounts for 0.376 of the variance. Second is the eigenvector with an eigenvalue of 0.8, which accounts for 0.334 of variance bringing the total up to 0.710 of variance accounted for.
>
> Thus the eigenvectors selected to account for 70% of the variance are the vectors whose eigenvalues are 0.9 and 0.8 respectively.

3

First we need to find k PC's such that 90% of the variance is accounted for. $k = 1, \frac{v}{2} \to k = 2, \frac{3v}{4} \to k = 3, \frac{7v}{8} \to k = 4, \frac{15v}{16}$. Thus PCA produces a dimensionality of 4 in order to have 90% of variance accounted for. Since in LDA, the number of dimensions is equal to the number of class labels minus 1 (ie $C - 1$), thus the number of class labels is 5 because 5 class labels minus 1 results in our desired 4 dimensions.

4

$C = 2$ from $|y| = 2$ where $y \in \{0, 1\}$

$D_0 = \{x^1, x^2, x^3, x^4\}$

$D_1 = \{x^5, x^6, x^7, x^8\}$

m_i represents the mean of all points in the class i

> $m_0 = [-2, 0]$
>
> $m_1 = [2, 0]$

$S_i = \sum_{x \in D_i} (x - m_i)(x - m_i)^T$

> $S_0 = [-1, 0][-1, 0]^T + [1, 0][1, 0]^T + [0, 1][0, 1]^T + [0, -1][0, -1]^T$

$$S_0 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$S_1 = [1,0][1,0]^T + [-1,0][-1,0]^T + [0,1][0,1]^T + [0,-1][0,-1]^T$$

$$S_1 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$S_W = \sum_{i=0}^{1} \frac{S_i}{|D_i|}$

$$S_W = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

m represents the mean of all points in our dataset

$m = 0, 0$

$S_B = \sum_{i=0}^{1} |D_i|(m_i - m)(m_i - m)^T$

$S_B = 4[-2,0][-2,0]^T + 4[2,0][2,0]^T$

$$S_B = \begin{pmatrix} 32 & 0 \\ 0 & 0 \end{pmatrix}$$

5

The curve shows a model which performs well on training data but performs significantly worse on unseen data. This means the model might be subject to overfitting. As such we should increase the penalization term, ie increase λ in order to potentially improve our model. Since the starting model had the L2 regularization term with $\lambda = 1$, we should try it with $\lambda = 2$ to see if this improves our model.

6

$|\gamma| = 5, |C| = 5$ so grid search results in $5 \times 5 = 25$ models trained. Then it is given that there were 10 models trained by the randomized search algorithm. Thus this results in a reduction of computation of $25/10 = 2.5$.

7

a

The accuracies for the models are shown in the table below:

Model Type	Accuracy
No Penalty	99.077%
L1, C=0.00001	76.308%
L2, C=0.00001	76.308%
L1, C=0.01	98.462%
L2, C=0.01	98.821%
L1, C=1	99.026%
L2, C=1	99.026%

b

The logistic regression without any penalty term resulted in a L2 norm of 81.93 which is found by finding the squared sum of all the weights, ie $\sum w^2$

c

The logistic regression with the L1 penalty term resulted in a L2 norm of 36.34 which is found by finding the squared sum of all the weights, ie $\sum w^2$. This is significantly lower than the L2 norm associated with the no penalty term logistic regression.

d

The logistic regression with the L2 penalty term resulted in a L2 norm of 37.35 which is found by finding the squared sum of all the weights, ie $\sum w^2$. This is significantly lower than the L2 norm associated with the no penalty term logistic regression.

e

The weights of the no penalty model were: [0.164, -1.288, 0.409, 5.424, -0.912, -1.389, 3.818, -5.332, 0.145, -0.796, -2.044, -0.248]

The weights of the best L1 penalty model were: [-0.598, -1.435, 0.272, 3.169, -0.957, -0.811, 3.361, -2.99, -0.46, -0.971, -0.895, -0.201]

The weights of the best L2 penalty model were: [-0.523, -1.366, 0.32, 3.351, -0.96, -0.747, 3.172, -3.195, -0.393, -0.917, -1.044, -0.222]

Thus according to these weights from the models, none of the models had any weights that were exactly 0. It is important to note that these models were trained on standardized data which would contribute to the size of the weights.

8

1

The number of dimensions in the dataset excluding the style column is $d = 12$

4

$$\lambda_1 = 3.042, \lambda_2 = 2.65, \lambda_3 = 1.642, \lambda_4 = 1.069, \lambda_5 = 0.841, \lambda_6 = 0.661$$

$$\lambda_7 = 0.564, \lambda_8 = 0.516, \lambda_9 = 0.458, \lambda_10 = 0.299, \lambda_11 = 0.228, \lambda_12 = 0.033$$

$$v_1 = \begin{pmatrix} 0.257 \\ 0.262 \\ -0.467 \\ -0.144 \\ 0.165 \\ -0.03 \\ -0.393 \\ -0.001 \\ 0.424 \\ -0.272 \\ 0.277 \\ -0.335 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0.395 \\ 0.105 \\ 0.28 \\ -0.08 \\ 0.148 \\ 0.383 \\ -0.445 \\ -0.31 \\ -0.123 \\ 0.494 \\ -0.141 \\ -0.082 \end{pmatrix}, \quad v_3 = \begin{pmatrix} -0.146 \\ 0.144 \\ -0.588 \\ 0.056 \\ -0.235 \\ -0.362 \\ -0.048 \\ -0.445 \\ -0.246 \\ 0.33 \\ -0.229 \\ 0.001 \end{pmatrix}, \quad v_4 = \begin{pmatrix} -0.319 \\ 0.343 \\ 0.076 \\ 0.112 \\ 0.508 \\ 0.063 \\ 0.096 \\ -0.082 \\ -0.488 \\ -0.207 \\ -0.005 \\ -0.451 \end{pmatrix},$$

$$v_5 = \begin{pmatrix} 0.313 \\ 0.27 \\ -0.047 \\ 0.165 \\ -0.394 \\ 0.425 \\ 0.473 \\ -0.376 \\ -0.044 \\ -0.239 \\ 0.193 \\ -0.043 \end{pmatrix}, \quad v_6 = \begin{pmatrix} -0.423 \\ 0.111 \\ 0.099 \\ 0.303 \\ -0.248 \\ 0.283 \\ -0.363 \\ -0.12 \\ 0.301 \\ -0.303 \\ -0.486 \\ -0.001 \end{pmatrix}, v_7 = \begin{pmatrix} -0.474 \\ 0.144 \\ 0.101 \\ 0.132 \\ -0.224 \\ 0.107 \\ -0.235 \\ -0.011 \\ 0.002 \\ 0.295 \\ 0.72 \\ 0.064 \end{pmatrix}, \quad v_8 = \begin{pmatrix} 0.092 \\ 0.555 \\ 0.052 \\ 0.151 \\ 0.33 \\ -0.155 \\ -0.013 \\ -0.043 \\ 0.071 \\ -0.077 \\ 0.003 \\ 0.716 \end{pmatrix},$$

$$v_9 = \begin{pmatrix} 0.208 \\ -0.153 \\ 0.407 \\ 0.471 \\ -0.001 \\ -0.561 \\ -0.079 \\ -0.362 \\ 0.137 \\ -0.112 \\ 0.139 \\ -0.207 \end{pmatrix}, \quad v_{10} = \begin{pmatrix} 0.3 \\ 0.12 \\ -0.169 \\ 0.588 \\ -0.193 \\ 0.02 \\ -0.17 \\ 0.592 \\ -0.297 \\ 0.085 \\ -0.047 \\ -0.078 \end{pmatrix}, \quad v_{11} = \begin{pmatrix} 0.059 \\ -0.493 \\ -0.213 \\ 0.08 \\ 0.116 \\ 0.169 \\ -0.339 \\ -0.226 \\ -0.417 \\ -0.416 \\ 0.191 \\ 0.332 \end{pmatrix}, \quad v_{12} = \begin{pmatrix} -0.087 \\ -0.297 \\ -0.296 \\ 0.472 \\ 0.459 \\ 0.278 \\ 0.273 \\ -0.093 \\ 0.357 \\ 0.308 \\ 0.018 \\ 0.008 \end{pmatrix}.$$

5

Thus the top 6 eigenvectors and eigenvalues are $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_6$ for the eigenvalues and $v_1, v_2, v_3, v_4, v_5, v_6$ for the eigenvectors. Note that v_i denotes the ith column of v.

6

Thus it follows that our projection matrix W is defined as:

$$W = (v_1, v_2, v_3, v_4, v_5, v_6) = \begin{pmatrix} 0.257 & 0.395 & -0.146 & -0.319 & 0.313 & -0.423 \\ 0.262 & 0.105 & 0.144 & 0.343 & 0.27 & 0.111 \\ -0.467 & 0.28 & -0.588 & 0.076 & -0.047 & 0.099 \\ -0.144 & -0.08 & 0.056 & 0.112 & 0.165 & 0.303 \\ 0.165 & 0.148 & -0.235 & 0.508 & -0.394 & -0.248 \\ -0.03 & 0.383 & -0.362 & 0.063 & 0.425 & 0.283 \\ -0.393 & -0.445 & -0.048 & 0.096 & 0.473 & -0.363 \\ -0.001 & -0.31 & -0.445 & -0.082 & -0.376 & -0.12 \\ 0.424 & -0.123 & -0.246 & -0.488 & -0.044 & 0.301 \\ -0.272 & 0.494 & 0.33 & -0.207 & -0.239 & -0.303 \\ 0.277 & -0.141 & -0.229 & -0.005 & 0.193 & -0.486 \\ -0.335 & -0.082 & 0.001 & -0.451 & -0.043 & -0.001 \end{pmatrix}$$

9

2

For classes *red* and *white*:

$$m_{red} = \begin{pmatrix} 0.85188711 \\ 1.14293625 \\ -0.32797945 \\ -0.61050298 \\ 0.89728426 \\ -0.82546595 \\ -1.22575798 \\ 0.68370347 \\ 0.57603762 \\ 0.85272394 \\ -0.05770297 \\ -0.208838 \end{pmatrix}, m_{white} = \begin{pmatrix} 0.27810688 \\ -0.37312272 \\ 0.1070721 \\ 0.19930467 \\ -0.29292722 \\ 0.26948143 \\ 0.40016068 \\ -0.22320168 \\ -0.18805311 \\ -0.27838007 \\ 0.0188377 \\ 0.0681773 \end{pmatrix}$$

3

```
 Between-class scatter matrix S_B:
[[ 1539.2410698   2065.12623927  -592.61307846 -1103.09365673
   1621.26738435 -1491.50172535 -2214.77353185  1235.35671034
   1040.81955712  1540.75310884  -104.26121538  -377.34183273]
```

```
[ 2065.12623927  2770.68125833  -795.08066805 -1479.9680827
  2175.17702846 -2001.07664055 -2971.4558846   1657.41910572
  1396.41789706  2067.15486985  -139.88229385  -506.2615176 ]
[ -592.61307846  -795.08066805   228.15806286   424.69483213
  -624.19349022   574.23326751   852.69538772  -475.61656031
  -400.71909071  -593.19521867    40.14092466   145.27789669]
[-1103.09365673 -1479.9680827    424.69483213   790.5295924
 -1161.87763088  1068.88136272  1587.21247893  -885.31561281
  -745.90099873 -1104.17725611    74.71856591   270.42117721]
[ 1621.26738435  2175.17702846  -624.19349022 -1161.87763088
  1707.66488961 -1570.98400534 -2332.79904061  1301.18899619
  1096.28493811  1622.86000011  -109.81730625  -397.45041771]
[-1491.50172535 -2001.07664055   574.23326751  1068.88136272
 -1570.98400534  1445.24300993  2146.08264347 -1197.0422964
 -1008.53868551 -1492.96686872   101.02756851   365.63862907]
[-2214.77353185 -2971.4558846    852.69538772  1587.21247893
 -2332.79904061  2146.08264347  3186.77944191 -1777.52231157
 -1497.60791325 -2216.9491651    150.01872336   542.94724848]
[ 1235.35671034  1657.41910572  -475.61656031  -885.31561281
  1301.18899619 -1197.0422964  -1777.52231157   991.46665959
   835.33596483  1236.57023538   -83.67746585  -302.84519709]
[ 1040.81955712  1396.41789706  -400.71909071  -745.90099873
  1096.28493811 -1008.53868551 -1497.60791325   835.33596483
   703.79186973  1041.84198293   -70.50040059  -255.15480774]
[ 1540.75310884  2067.15486985  -593.19521867 -1104.17725611
  1622.86000011 -1492.96686872 -2216.9491651   1236.57023538
  1041.84198293  1542.2666332   -104.36363406  -377.71250604]
[ -104.26121538  -139.88229385    40.14092466    74.71856591
  -109.81730625   101.02756851   150.01872336   -83.67746585
   -70.50040059  -104.36363406     7.0621823     25.55942592]
[ -377.34183273  -506.2615176    145.27789669   270.42117721
  -397.45041771   365.63862907   542.94724848  -302.84519709
  -255.15480774  -377.71250604    25.55942592    92.50458653]]

  Within-class scatter matrix S_W:
[[ 4.95775893e+03 -6.42229598e+02  2.70047199e+03  3.75551274e+02
   3.16104047e+02 -3.45430353e+02  7.69103351e+01  1.74618144e+03
  -2.68261450e+03  4.05538526e+02 -5.15887327e+02 -1.21258789e+02]
 [-6.42229598e+02  3.72631874e+03 -1.66066395e+03  2.06483483e+02
   2.74999396e+02 -2.89488179e+02  2.78604048e+02  1.05188718e+02
   3.02251358e+02 -5.98938903e+02 -1.04667293e+02 -1.21998799e+03]
 [ 2.70047199e+03 -1.66066395e+03  6.26884194e+03  5.00810783e+02
   8.77563588e+02  2.90685117e+02  4.15791730e+02  1.10032864e+03
```

```
 -1.74204473e+03  9.58309078e+02 -1.08317143e+02  4.10421670e+02]
[ 3.75551274e+02  2.06483483e+02  5.00810783e+02  5.70647041e+03
  3.24151203e+02  1.54856919e+03  1.63193139e+03  4.47501824e+03
 -9.90875981e+02 -1.03793096e+02 -2.40983633e+03 -5.10683386e+02]
[ 3.16104047e+02  2.74999396e+02  8.77563588e+02  3.24151203e+02
  4.78933511e+03  3.03778036e+02  5.16040024e+02  1.05471843e+03
 -8.05817195e+02  9.47309713e+02 -1.55936322e+03 -9.06273339e+02]
[-3.45430353e+02 -2.89488179e+02  2.90685117e+02  1.54856919e+03
  3.03778036e+02  5.05175699e+03  2.53782608e+03  1.36412462e+03
  6.09259206e+01  2.68560123e+02 -1.26943788e+03 -5.29513723e+00]
[ 7.69103351e+01  2.78604048e+02  4.15791730e+02  1.63193139e+03
  5.16040024e+02  2.53782608e+03  3.31022056e+03  1.98798946e+03
 -5.13620163e+01  4.25552016e+02 -1.87652916e+03 -8.11828542e+02]
[ 1.74618144e+03  1.05188718e+02  1.10032864e+03  4.47501824e+03
  1.05471843e+03  1.36412462e+03  1.98798946e+03  5.50553334e+03
 -7.59411499e+02  4.49261549e+02 -4.37810754e+03 -1.68431362e+03]
[-2.68261450e+03  3.02251358e+02 -1.74204473e+03 -9.90875981e+02
 -8.05817195e+02  6.09259206e+01 -5.13620163e+01 -7.59411499e+02
  5.79320813e+03  2.06383790e+02  8.58251691e+02  3.81883365e+02]
[ 4.05538526e+02 -5.98938903e+02  9.58309078e+02 -1.03793096e+02
  9.47309713e+02  2.68560123e+02  4.25552016e+02  4.49261549e+02
  2.06383790e+02  4.95473337e+03  8.46829545e+01  6.27752448e+02]
[-5.15887327e+02 -1.04667293e+02 -1.08317143e+02 -2.40983633e+03
 -1.55936322e+03 -1.26943788e+03 -1.87652916e+03 -4.37810754e+03
  8.58251691e+02  8.46829545e+01  6.48993782e+03  2.86117800e+03]
[-1.21258789e+02 -1.21998799e+03  4.10421670e+02 -5.10683386e+02
 -9.06273339e+02 -5.29513723e+00 -8.11828542e+02 -1.68431362e+03
  3.81883365e+02  6.27752448e+02  2.86117800e+03  6.40449541e+03]]
```

4

```
Eigenvalues of S_W^{-1} S_B:
  1 = 0.0
  2 = 6.2696317063212685
  3 = -7.656543900203237e-16
  4 = 3.8552904845963943e-16
  5 = 5.2521240269199517e-17
  6 = 5.2521240269199517e-17
  7 = 4.656785883267749e-17
  8 = 4.656785883267749e-17
  9 = -4.8154770905149857e-17
```

```
  10 = -4.8154770905149857e-17
  11 = -4.222787296529956e-17
  12 = -2.8089972508736594e-17

  Corresponding eigenvectors (each column):
[[-8.38330887e-01  1.17920265e-01 -2.57760409e-01 -3.94115006e-01
  -1.57146273e-01 -1.57146273e-01  7.25389030e-01  7.25389030e-01
  -8.45590914e-01 -8.45590914e-01 -7.36422774e-01  6.53458524e-01]
 [ 7.97805804e-02 -1.45321930e-01  4.05811333e-01 -3.02636112e-01
   1.56905874e-01  1.56905874e-01 -8.74553608e-02 -8.74553608e-02
   8.14752592e-02  8.14752592e-02 -3.23824208e-02 -1.44030399e-01]
 [-1.88941980e-02  3.44161613e-02  2.47361421e-02 -4.27994800e-03
  -5.90877600e-02 -5.90877600e-02 -5.92913438e-02 -5.92913438e-02
  -3.06941007e-02 -3.06941007e-02  1.32560246e-04  3.98184086e-02]
 [-2.55550612e-01  4.65490575e-01  2.65226476e-01 -3.22379722e-02
  -6.65168387e-01 -6.65168387e-01  2.84805222e-01  2.84805222e-01
  -2.02460135e-01 -2.02460135e-01 -2.33047669e-01  1.03979429e-01]
 [ 2.71748087e-02 -4.94994601e-02  7.20273024e-02 -1.90702892e-02
  -1.92183863e-01 -1.92183863e-01  6.03478498e-03  6.03478498e-03
   2.37520083e-02  2.37520083e-02  6.61540967e-02  9.45511015e-03]
 [ 4.95742002e-02 -9.03004019e-02  1.83571376e-01  4.18960058e-02
  -9.16287774e-03 -9.16287774e-03 -4.03899656e-02 -4.03899656e-02
   5.86840098e-02  5.86840098e-02  9.06947730e-02 -2.00305252e-03]
 [-1.68559728e-01  3.07034932e-01 -3.26279123e-01 -3.03310879e-01
   1.15982528e-01  1.15982528e-01  1.45363061e-01  1.45363061e-01
  -1.57172891e-01 -1.57172891e-01 -1.24001347e-01  1.30627864e-01]
 [ 4.14530525e-01 -7.55075681e-01 -2.34404940e-01  6.34041929e-01
  -3.21334950e-01 -3.21334950e-01 -3.95646919e-01 -3.95646919e-01
   3.97326719e-01  3.97326719e-01  1.95281882e-01 -4.28348819e-01]
 [-2.80468744e-02  5.10879452e-02 -3.06884433e-01 -1.63544816e-01
   5.14490814e-02  5.14490814e-02 -1.23912614e-02 -1.23912614e-02
  -4.17090600e-02 -4.17090600e-02 -1.23756274e-01 -3.68141951e-02]
 [ 1.74194333e-02 -3.17298477e-02 -1.22014974e-02  4.42009205e-02
   2.66667379e-02  2.66667379e-02  5.05076807e-02  5.05076807e-02
   8.28827050e-02  8.28827050e-02  2.93189912e-01  1.16256470e-01]
 [ 1.42855278e-01 -2.60213760e-01  6.34881359e-01  4.74646082e-01
  -3.81220675e-02 -3.81220675e-02 -3.43612881e-01 -3.43612881e-01
   8.96177099e-02  8.96177099e-02 -4.24347623e-01 -5.63749858e-01]
 [ 1.52175255e-02 -2.77190285e-02  1.51121516e-02  4.37264459e-02
  -5.93342034e-03 -5.93342034e-03  4.49854220e-02  4.49854220e-02
  -7.13389797e-02 -7.13389797e-02 -2.34225413e-01 -8.07803167e-02]]
```

5

```
 Sorted eigenvalues (descending): [ 6.26963171e+00  3.85529048e-16  5.25212403e-
17  5.25212403e-17
  4.65678588e-17  4.65678588e-17  0.00000000e+00 -2.80899725e-17
 -4.22278730e-17 -4.81547709e-17 -4.81547709e-17 -7.65654390e-16]
   Selected k = 6 largest eigenvalues: [6.26963171e+00 3.85529048e-16 5.25212403e-
17 5.25212403e-17
 4.65678588e-17 4.65678588e-17]
   Corresponding eigenvectors (columns):
[[ 0.11792026 -0.39411501 -0.15714627 -0.15714627  0.72538903  0.72538903]
 [-0.14532193 -0.30263611  0.15690587  0.15690587 -0.08745536 -0.08745536]
 [ 0.03441616 -0.00427995 -0.05908776 -0.05908776 -0.05929134 -0.05929134]
 [ 0.46549058 -0.03223797 -0.66516839 -0.66516839  0.28480522  0.28480522]
 [-0.04949946 -0.01907029 -0.19218386 -0.19218386  0.00603478  0.00603478]
 [-0.0903004   0.04189601 -0.00916288 -0.00916288 -0.04038997 -0.04038997]
 [ 0.30703493 -0.30331088  0.11598253  0.11598253  0.14536306  0.14536306]
 [-0.75507568  0.63404193 -0.32133495 -0.32133495 -0.39564692 -0.39564692]
 [ 0.05108795 -0.16354482  0.05144908  0.05144908 -0.01239126 -0.01239126]
 [-0.03172985  0.04420092  0.02666674  0.02666674  0.05050768  0.05050768]
 [-0.26021376  0.47464608 -0.03812207 -0.03812207 -0.34361288 -0.34361288]
 [-0.02771903  0.04372645 -0.00593342 -0.00593342  0.04498542  0.04498542]]
```

6

```
 Projection matrix W (each column is an eigenvector):
[[ 0.11792026 -0.39411501 -0.15714627 -0.15714627  0.72538903  0.72538903]
 [-0.14532193 -0.30263611  0.15690587  0.15690587 -0.08745536 -0.08745536]
 [ 0.03441616 -0.00427995 -0.05908776 -0.05908776 -0.05929134 -0.05929134]
 [ 0.46549058 -0.03223797 -0.66516839 -0.66516839  0.28480522  0.28480522]
 [-0.04949946 -0.01907029 -0.19218386 -0.19218386  0.00603478  0.00603478]
 [-0.0903004   0.04189601 -0.00916288 -0.00916288 -0.04038997 -0.04038997]
 [ 0.30703493 -0.30331088  0.11598253  0.11598253  0.14536306  0.14536306]
 [-0.75507568  0.63404193 -0.32133495 -0.32133495 -0.39564692 -0.39564692]
 [ 0.05108795 -0.16354482  0.05144908  0.05144908 -0.01239126 -0.01239126]
 [-0.03172985  0.04420092  0.02666674  0.02666674  0.05050768  0.05050768]
 [-0.26021376  0.47464608 -0.03812207 -0.03812207 -0.34361288 -0.34361288]
 [-0.02771903  0.04372645 -0.00593342 -0.00593342  0.04498542  0.04498542]]
```